

Sustainable and intelligent additive manufacturing

Early detection of production errors in 3D printing using artificial intelligence

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Sustainable and Intelligent Additive Manufacturing – Early Recognition of Manufacturing Defects in 3D Printing with Artificial Intelligence

Additive manufacturing is an increasingly important manufacturing technology with huge economic potential. However, its popularity is accompanied by high material and time losses, as defects are often detected at a very late stage. One solution for more sustainable production is the automated detection of manufacturing defects using artificial intelligence. This article describes the digitization of the defect detection process in additive manufacturing using a system based on a neural network. In addition to the steps for automated defect detection, system performance is also discussed.

Keywords:

additive manufacturing, artificial neural networks, computer vision, automation

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Additive manufacturing is an increasingly important manufacturing technology with great economic potential. However, its popularity is accompanied by high material and time losses, as defective workpieces are often only detected very late in the manufacturing process. One approach to more sustainable and efficient production is the automated and early detection of manufacturing defects using artificial intelligence methods. This article describes the digitization of the error detection process in 3D printing using an image-based learning system. In addition to the individual steps involved in automated error detection, the performance of the system is also discussed.

Additive manufacturing processes, also known as 3D printing, are widely used and find application in many different areas, from mechanical engineering [1] to special applications in biotechnology [2]. Due to its easy accessibility, the filament-based 3D printing process (fused filament fabrication, FFF) is the most commonly used production technology in additive manufacturing [3]. In the FFF process, components are additively manufactured from plastic, which is fed into the printing process in the form of wire-like filaments. The printed objects are built up layer by layer on a print bed by selectively depositing individual strands of molten plastic until the finished model is complete.

Additive manufacturing is generally considered a sustainable process, as almost only the amount of material required for the component needs to be used. However, manufacturing errors and support structures for the printing process also result in plastic waste in this process. Various errors can occur during the manufacturing process, which differ visually in their severity.

Serious printing errors include those that prevent the print object from being built completely, for example by clogging the extrusion nozzle or detaching the unfinished component from the print bed. In the case of serious printing errors, the component is unusable and is discarded as plastic waste. According to a study [4], 34% of the plastic used in the printing process ends up as plastic waste, 55% of which is due to faulty prints and 45% to support structures. Early detection of printing errors and situation-dependent termination of the printing process can reduce the amount of material used for misprints and thus significantly reduce the amount of plastic waste produced. However, early detection of printing errors is only possible through continuous monitoring of the printing process.

There are already several research approaches to monitoring the printing process using artificial intelligence (AI) methods, specifically machine vision approaches, to automatically detect printing errors [5, 6]. Commercial solutions for this are also available.

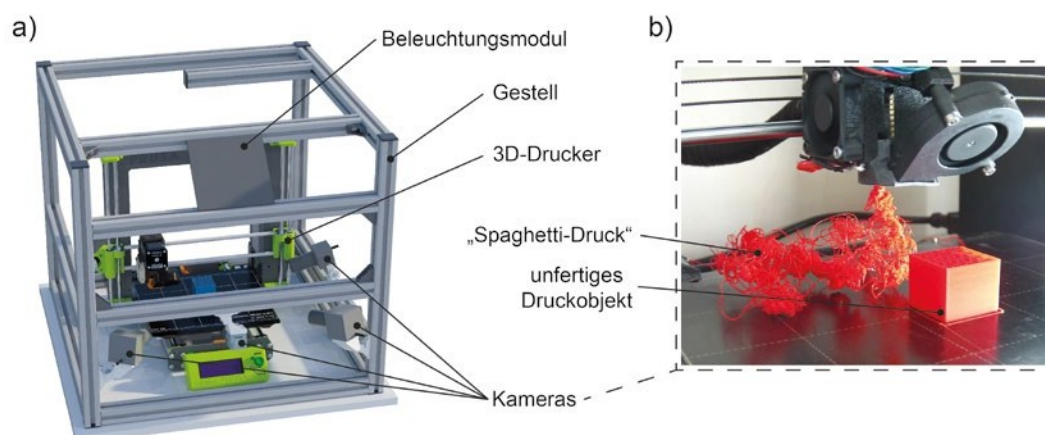


Figure 1: 3D printer with camera system for monitoring the printing process and detecting manufacturing errors (a) and image capture of "spaghetti print" (b) with the test setup.

monitoring the printing process are already in development [7, 8]. The autonomous methods for print monitoring are based on artificial neural networks that have been trained to detect manufacturing errors using image material. To monitor the printing process, images of the printing progress are continuously recorded and manufacturing errors are detected in real time using the trained neural networks. When printing errors are detected, appropriate steps can be taken, such as automatically pausing or canceling the print job.

The research project "Efficient, intelligent, generative manufacturing with recycled plastics through AI optimization (KI-GenF)" [9], funded by the Carl Zeiss Foundation, addresses the question of how sustainable production methods can be further developed using artificial intelligence processes. In addition to improving the sustainability of the FFF printing process and recycling plastic printing waste, AI-supported methods are used to detect printing errors at an early stage, thereby reducing the amount of printing waste.

This work deals with the detection of printing errors by monitoring the printing process with cameras and artificial neural networks. In contrast to previous work, several cameras are used here to capture the printing process simultaneously from different perspectives. First, the most serious printing errors are considered, as this is where the greatest potential for savings in plastic waste exists. In addition to generating and annotating suitable image material, the training process of a neural network is described and the performance of the system in error detection is recorded and evaluated. Finally, an outlook on future work is provided.

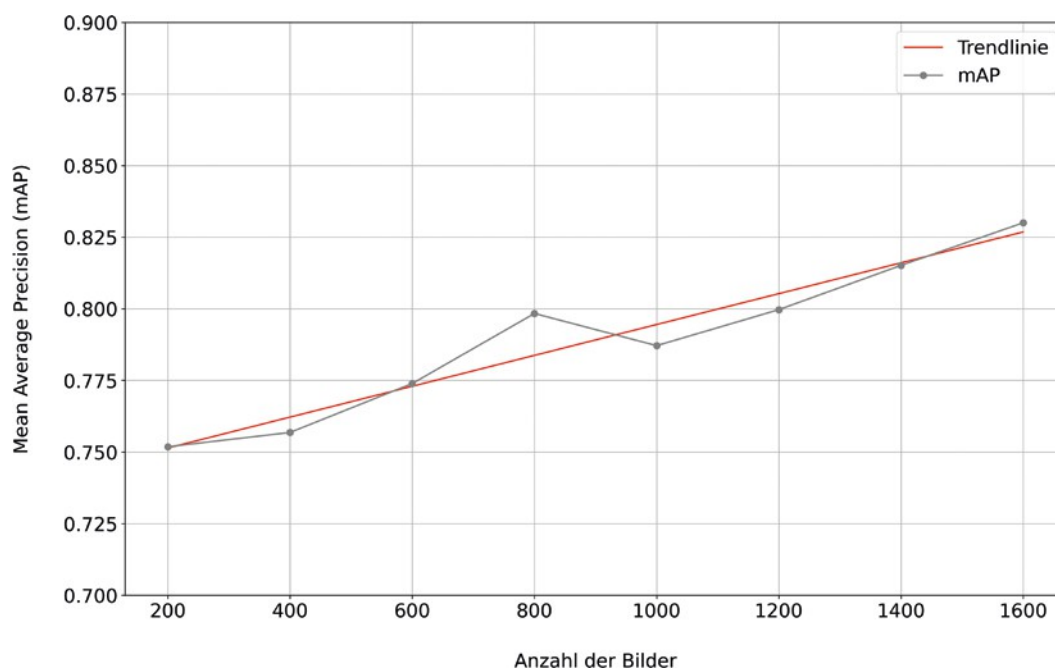
From manual to automated error detection

In many cases, printing errors in additive manufacturing are only detected at a very late stage. Typically, they are currently identified by the observer based on their visual characteristics, who then initiates manual countermeasures, such as canceling or pausing the print. The sustainable digitization of this process can minimize the use of materials and time. This requires methods from the field of image processing for the automated capture and storage of image data. Furthermore, methods are needed that can independently, accurately, and reliably extract the visual characteristics of the printed objects from digital images and assign them to a specific error category. Today, tasks of this kind are performed in many areas by artificial neural networks [10]. These are computer models for pattern recognition that learn patterns from annotated data and use this knowledge to automatically locate and classify image content. In this scientific paper, the YOLOv5 [11] model architecture is used to detect printing errors. Due to its speed and accuracy, it is currently one of the most popular network architectures in the field of object recognition.

Images of printing errors are generated in a targeted manner

A large number of images of misprints are required to achieve satisfactory accuracy in misprint detection. In order to generate the images in a time-efficient and reproducible manner, misprints were deliberately induced. To do this, the print files or the G-code used by the 3D printer to execute the printing process were manipulated.

Figure 2: Accuracy of the neural network with increasing training data volume. The trend line shows that the performance of the network increases with increasing training data volume



This resulted in a serious printing error known as "spaghetti printing" (Figure 1b). In "spaghetti printing," the plastic strands extruded by the printer suddenly can no longer be connected to the existing print object and accumulate as a curled structure below the extrusion nozzle. This behavior occurs in the printing process, for example, due to a sudden detachment of the print object from the print bed or a temporary interruption of the extrusion. A sudden detachment of the print object can occur, for example, due to a contaminated print bed or inappropriate print settings. The detachment of the print object is simulated by moving the print head in the print plane. The print head is moved to an area that makes it impossible for it to fuse with the underlying layers of the print object, achieving the same effect as a sudden detachment of the component. Extrusion skips can be caused by temporary clogging of the extrusion nozzle or problems with the filament feed and can be simulated by offsetting the print head in the height axis.

By manipulating the print files, misprints were generated for various basic geometries (cube, cylinder, pyramid) and images of the print error were taken from different angles using cameras. An FFF 3D printer (Prusa i3 MK3s, Prusa Research) and several cameras mounted on a frame were used to generate the print errors.

This made it possible to generate a sufficient number of images for training with little time and a minimal number of misprints. A lighting module attached to the frame ensured uniform illumination of the printing space. In addition to generating images for training the neural networks, the setup was also used to test print monitoring.

High data quality through careful annotation

Supervised learning is used for the early detection of printing errors, in which the neural network is trained with annotated image data that has already been assigned to the correct error category. After the image material has been generated, it must be prepared for training the neural network. Here, the printing error to be detected is marked on each image with a tightly enclosed bounding box. Each instance of the visible printing error must be marked, as partial marking can lead to poorer performance in subsequent object recognition. During the annotation process, each bounding box is assigned to the corresponding error category. A text file is then generated for each individual image, containing the name of the error and the position and size of the bounding boxes.

The printer was placed on a frame made of aluminum profiles (Figure 1a). A total of four cameras were used to simultaneously capture the printing errors from different perspectives.

For training the neural network used here, an image set of at least 1500 annotated individual images per error category is recommended [12]. In this work

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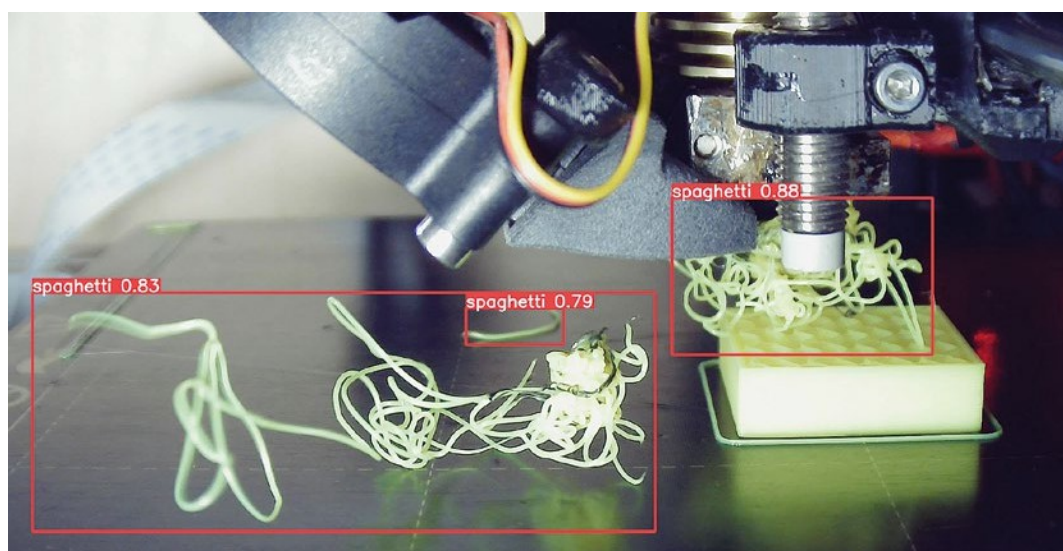


Figure 3: Image of a 3D print after automated print error detection with three instances of a print error found and probability values between 79% and 88%.

A total of 1,600 images were manually annotated for the automated detection of the error category "spaghetti printing" and thus prepared for training the artificial neural network. The graphical annotation tool Labellmg [13] was used for annotation.

Learning print error detection from image data

During the training process, the annotated image material passes through the neural network in several passes (epochs). Here, the weighted connections of the network are adjusted so that the correct error category can be predicted as accurately as possible for a printing error in an image. The training process is also known as deep learning.

[10]. The amount of data used was divided according to the 80/20 scheme. Here, 80% of the data is used for training, 10% of the data for validating the training process, and 10% of the data for testing the neural network. Furthermore, care was taken to ensure that 10% of the training data did not contain any printing errors, as this reduces the detection of false positive printing errors and increases the overall accuracy of the network. In addition, the number of epochs was set to 30, as a higher number of epochs resulted in overfitting of the model, which led to lower overall accuracy of the model.

Larger amounts of data lead to better results

To evaluate performance, the neural network was trained with different amounts of image data over 30 epochs, with the amount of image data gradually increased from 200 images

by 200 to a total of 1600 images. The performance metrics for each of the image data sets were then recorded. Figure 2 clearly shows that good results can already be achieved with small amounts of data. With a data set of 200 images, the accuracy of the neural network is already 75.2%. As the trend line shows, the performance of automated printing error detection can be improved with a larger and carefully annotated training data set. In this experiment, a mean average precision (mAP) of 83% was achieved in the detection of the printing error "spaghetti printing."

The trained network can be used to monitor printing processes in real time. The practical application of the trained network can be seen in Figure 3.

As the research project progresses, the plan is to transfer the presented method to the automated detection of other printing errors in additive manufacturing. Furthermore, dynamic adjustment of printing parameters will be used to intervene automatically and in real time in the printing process in order to reduce the occurrence of misprints and thus make additive manufacturing more sustainable.

This article was written as part of the research project "Efficient, intelligent, generative manufacturing with recycled plastics through AI optimization (KI-GenF)," which is funded by the Carl Zeiss Foundation with up to €1 million as part of the CZS Transfer funding program.

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